MILITARY SPECIFICATION SHEET

FIXTURES, LIGHTING; FLUORESCENT, DETAIL LIGHTING FOR RADIO OPERATOR'S DESK

8 WATTS, 120 VOLTS, 60 HERTZ, SYMBOLS 270.3 AND 270.4

This specification sheet is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the fixtures described herein shall consist of this document and the latest issue of MIL-F-16377(SHIPS).

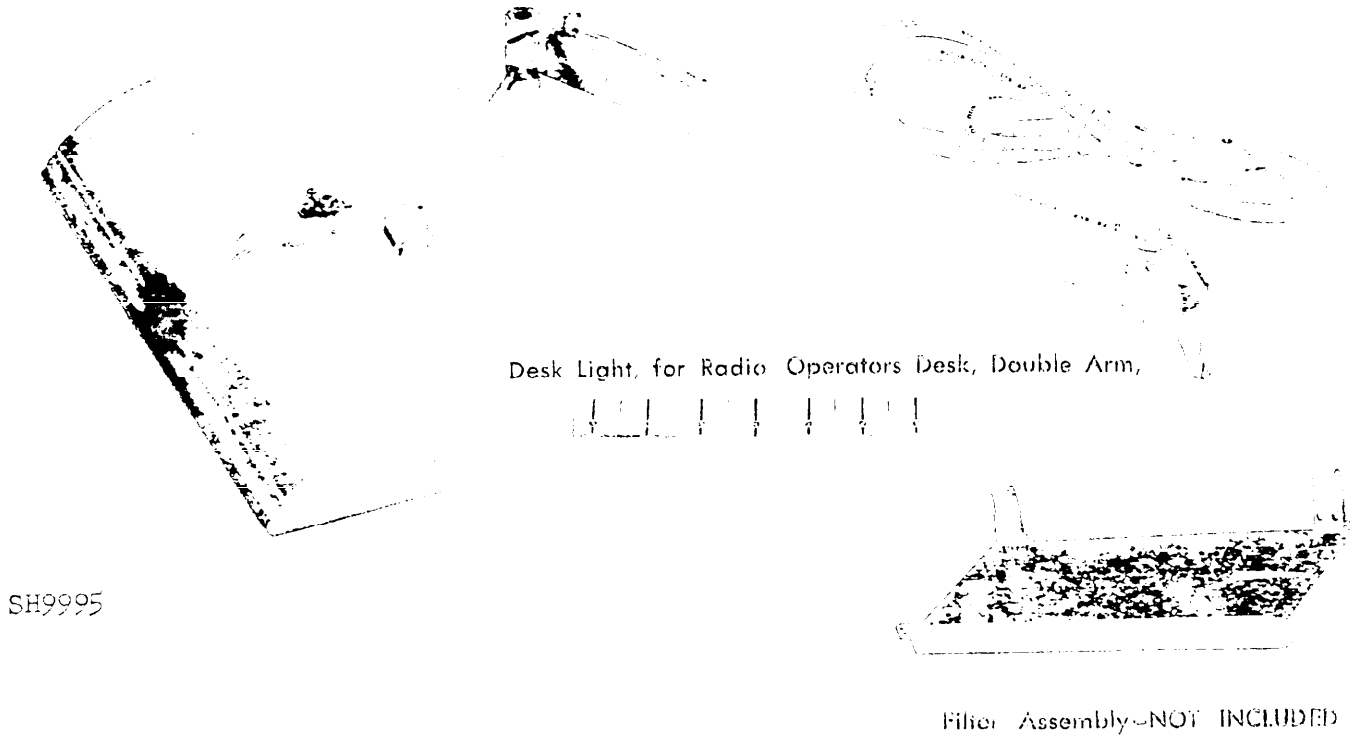
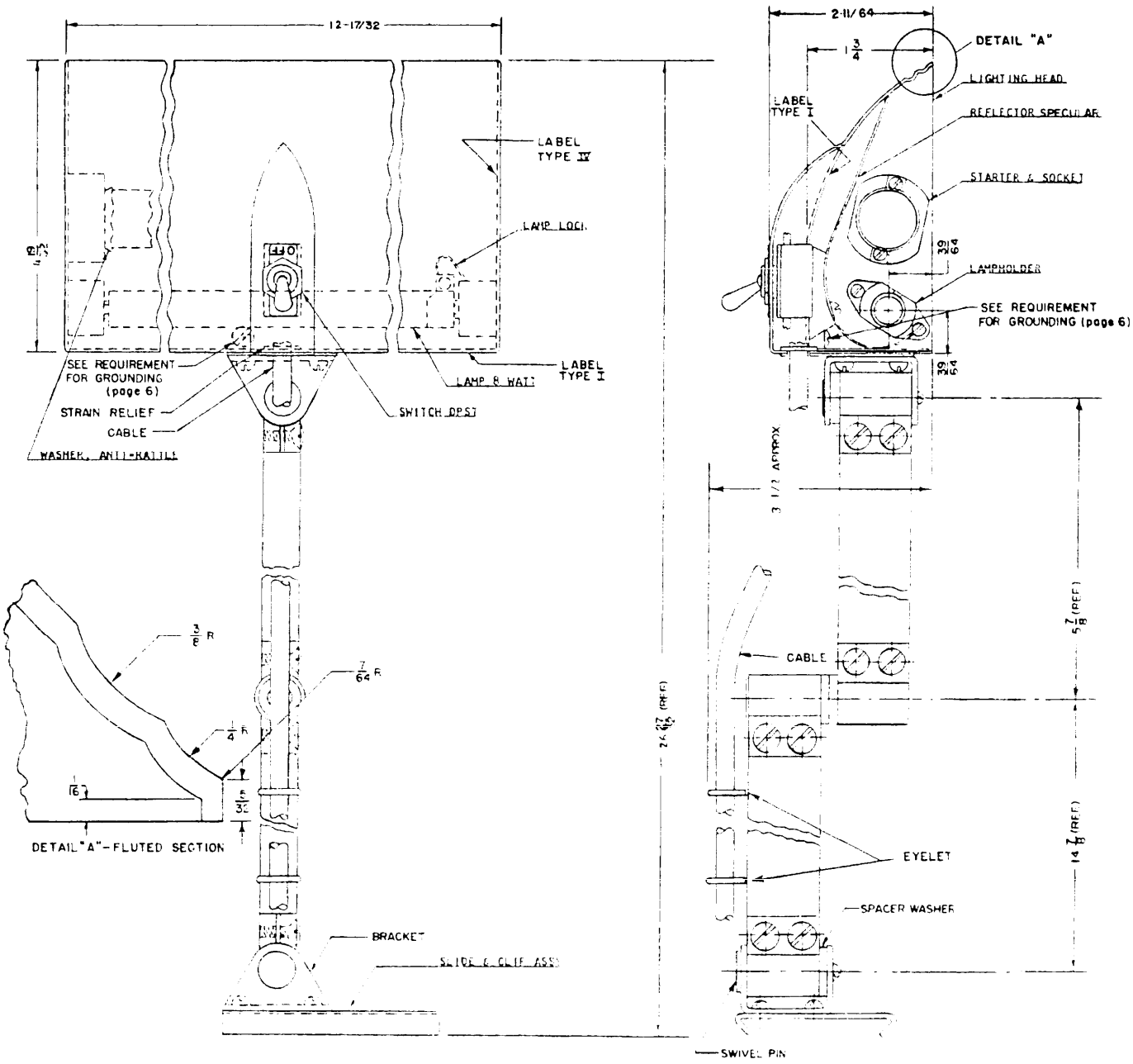


FIGURE 1. Typical view of Symbol 270.4.

All dimensions in inches.
For tolerances, other than specified herein, see paragraph 3.5.2 of MIL-F-16377.

FSC 6230

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FIGURE 2. Light, fluorescent, radio operators desk, double arm, Symbol 270.4.

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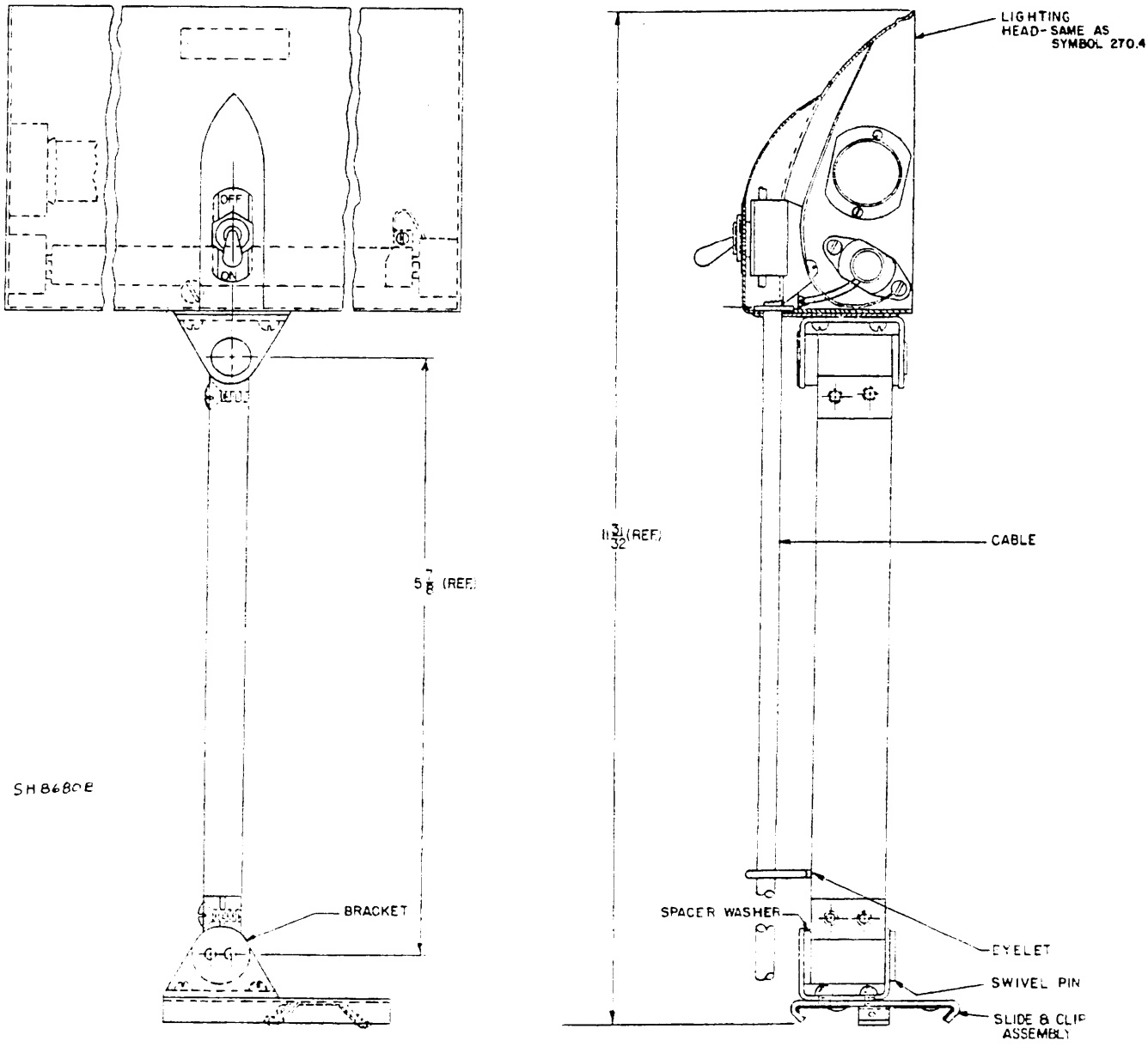


FIGURE 3. Light, fluorescent, radio operators desk, single arm, Symbol 270.3.

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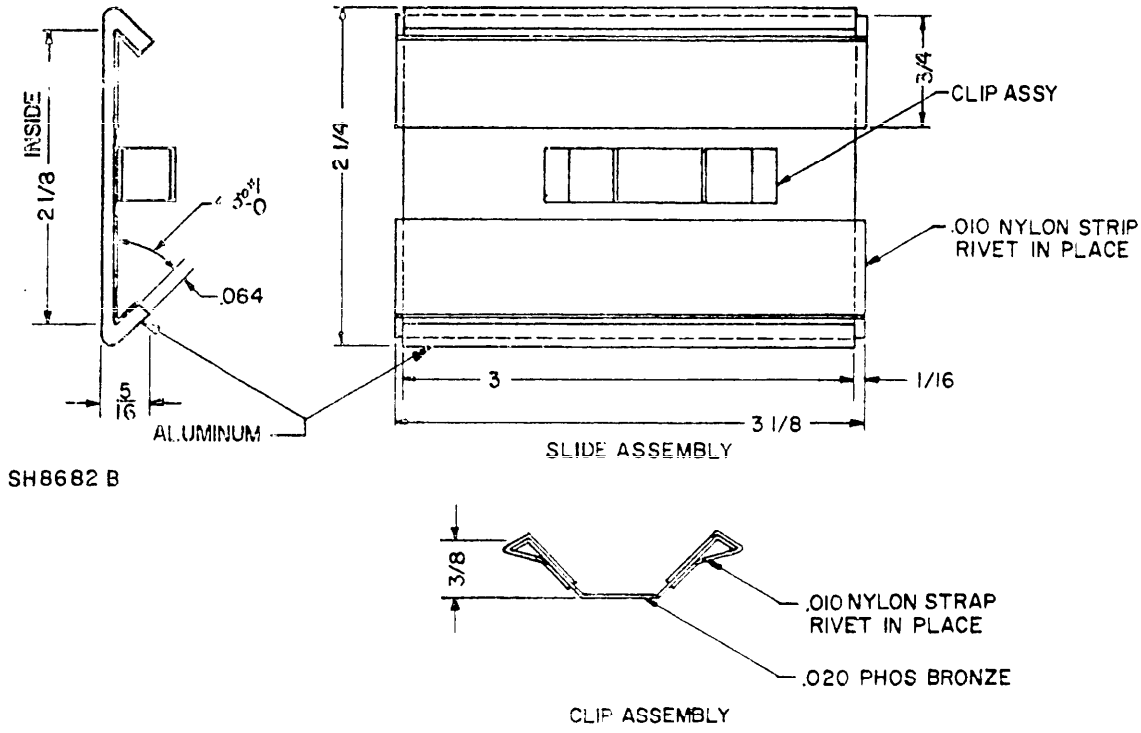


FIGURE 4. Slide and clip assemblies for Symbols 270.3 and 270.4.

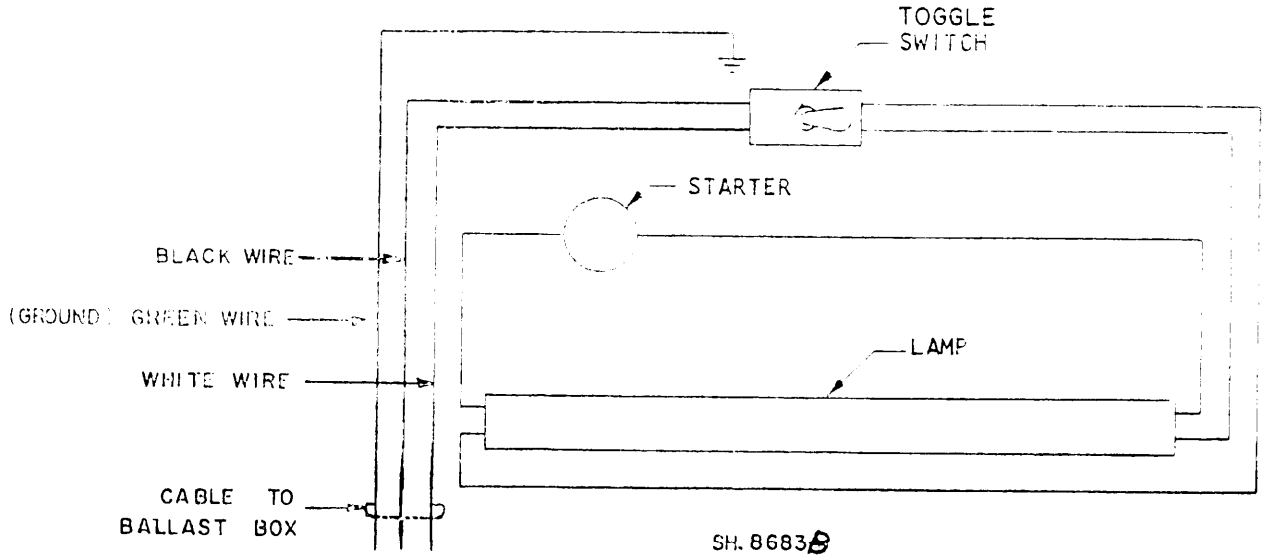
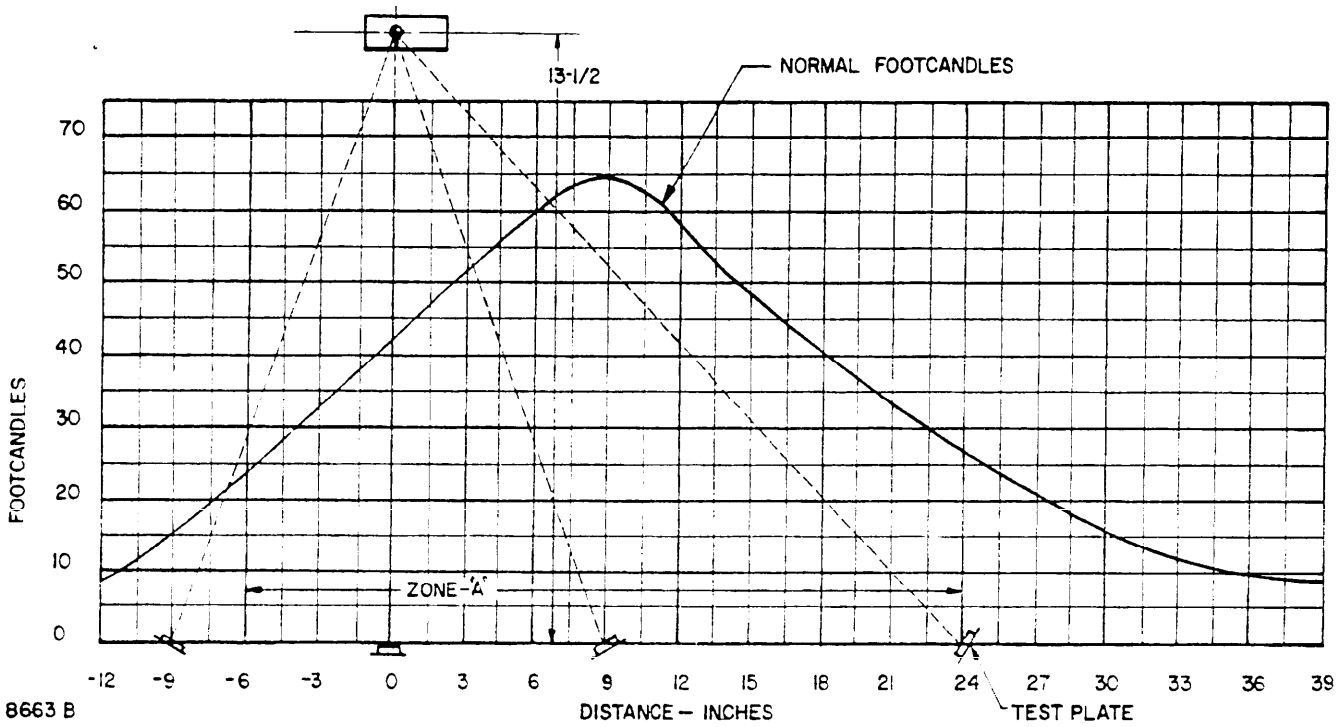


FIGURE 5. Wiring diagram for Symbols 270.3 and 270.4.

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FOOTCANDLE VALUES WITHIN ZONE "A" SHALL BE NOT LESS THAN 90 PERCENT OF THE VALUES SHOWN HEREON. THESE RESULTS SHALL BE MET AFTER CORRECTION TO 380 LUMENS OUTPUT FOR BOTH LAMP AND BALLAST HAS BEEN MADE.

Photometric test data:

Distance: 13½ inches.
Voltage: 115 volts, a.c., 1 phase.
Ambient temperature: 76°F.
Lamp and ballast furnished with fixture shall be used when testing.

FIGURE 6. Footcandle distribution curves for Symbols 270.3 and 270.4.

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REQUIREMENTS:

CLASSIFICATION:
Type: I
Class: 1

SYMBOL NUMBERS:

270.3: Light, single arm with one 8-watt cool white lamp.
270.4: Light, double arm, with one 8-watt cool white lamp.

DESIGN:
Electrical rating: 8 watts, 120 volts, 60 Hertz.
Dimensions and configuration: See figures 1, 2, 3 and 4.
Slide and clip assembly: See figure 4.
Reflector: Specular, prefinished lighting sheet.
Swivel pin: Nylon
Cable: 10 feet of SVT, construction in accordance with J-C-580 except with light gray thermoplastic sheath, and size 18 AWG, three conductor.
Weight:
Symbol 270.3: 1 pound 14 ounces maximum.
Symbol 270.4: 3 pounds 2 ounces maximum.
Strain Relief: Strain relief for cables shall be as shown on figures 2 and 3.
Distribution curves: Footcandle (see figure 6).
Wiring diagram: See figure 5.
Lighting head and supporting arm(s): The lighting head shall be made of 3/32 inch minimum thickness cast aluminum. The supporting arm(s) shall be made of 0.074 inch thickness extruded aluminum. The lighting head shall be designed to house the necessary parts, and shall be of a shape that will accommodate the filter assembly shown on MIL-F-16377/48, Symbol 121.1 and 121.2 without rattle or escapement of white light.
Grounding for lighting head only: For internal cable the green ground leads inside the lighting head assembly shall be terminated with crimp-type terminals and connected under round head machine screw and lockwasher.
Swivel joints: Swivel joints shall hold arm firm at all adjustments. Swivel action shall be smooth for all adjustments in the horizontal plane.
Instruction sheet: An instruction sheet containing the following information shall be packaged with each fixture:

"THIS FIXTURE IS DESIGNED FOR USE WITH A SEPARATELY MOUNTED 8-WATT BALLAST BOX SHOWN ON MIL-F-16377/46, SYMBOL 514. THE BALLAST BOX IS NOT FURNISHED WITH THE FIXTURE AND MUST BE ORDERED SEPARATELY. FOR INSTALLATION INSTRUCTIONS SEE NAVSEA 0964-LP-000-2000 COVERING 'LIGHTING ON NAVAL SHIPS'."

Designation and marking: The following labels shall be furnished in accordance with MIL-F-16377 applied as shown in figure 2.

(a) Two type I.
(b) One type IV. The information to be shown on the type IV label shall be "USE LAMP NSN 6240-00-229-5546 ONLY."

Standard Parts: The standard parts shown in table I shall be used in the construction of the fixtures specified herein.

QUALITY ASSURANCE: Quality assurance shall be as specified in MIL-F-16377 and as follows:

First article inspection	See table II
Quality conformance:	
Comparison inspection:	Same as the first article inspection.
Inspection of product for delivery:	
Examination:	See MIL-F-16377
Examination for preparation of delivery:	See MIL-F-16377
Quality conformance tests:	See MIL-F-16377

GENERAL INFORMATION: Assignment of Military part numbers and general information is as follows:

Military part number for Symbol 270.3: M16377/19-270.3.
Military part number for Symbol 270.4: M16377/19-270.4.
National stock number for Symbol 270.3: NSN 6230-00-548-0108
Formerly covered on NAVSHIPS Drawing 9000-S6401-73854 and MS17154.

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TABLE I. Standard parts.

Name	Specification or Military part number	National Stock Number (NSN)	Remarks
Lamp	W-L-116/3	6240-00-299-5546	Industry Number F8T5/CW
Lamp lock	M16377/40-001	6250-00-399-7039	
Lampholder	MIL-L-970/10	6250-00-377-8590	
Startersocket	MIL-L-970/13	6250-00-295-2738	
Starter	8W, type III of W-S-755	6250-00-299-5962	Commercial designation: FS5
Ballast	M16377/45-001	6250-00-295-2735	
Switch, toggle, DPST	MS16569-1	5930-00-719-9659	
Red and white filter	M16377/48-121.1	6230-00-548-2040	Not furnished
Blue and white filter	M16377/48-121.2		Not furnished

TABLE II. First article and comparison inspection.

Tests	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Shock	3.5.12	4.8.7	1/
Vibration	3.5.13	4.8.8	
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Noise	3.5.14	4.8.9	
Grounding circuit	3.6.13.3	4.8.11	
Light output	3.7.2	4.8.6.2	Footcandle
Magnetic permeability	3.4.8	4.8.18	

1/ Lamp breakage or lamp fallout during this test does not constitute failure.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6230-268)

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